Chapter 4 Breakout (Pt. III)

STAT 4714

Questions

1. Suppose that the probabilities that 0, 1, 2, or 3 power failures will strike a certain subdivision in any given year are 0.4, 0.3, 0.2, and 0.1, respectively. Find the mean and variance of the random variable X representing the number of power failures striking this particular subdivision. Find the mean and variance of the random variable $Z = 5X + 3$.

2. If a random variable X is defined such that $E[(X-1)^2] = 10$ and $E[(X-2)^2] = 6$, find μ and σ^2.

3. If X and Y are independent random variables with variances $\sigma_X^2 = 5$ and $\sigma_Y^2 = 3$, find the variance of the random variable $Z = -2X + 4Y - 3$.

4. Let X represent the number that occurs when a red die is tossed and Y denote the number that occurs when a green die is tossed. Find

   (a) $E(X+Y)$,

   (b) $E(X-Y)$,

   (c) $E(XY)$.

5. A random variable X has a mean $\mu = 10$ and a variance $\sigma^2 = 4$. Using Chebyshev's theorem, find

   (a) $P(|X-10| \geq 3)$,

   (b) $P(|X-10| < 3)$,

   (c) $P(5 < X < 15)$,

1. Suppose that the probabilities that 0, 1, 2, or 3 power failures will strike a certain subdivision in any given year are 0.4, 0.3, 0.2, and 0.1, respectively. Find the mean and variance of the random variable X representing the number of power failures striking this particular subdivision. Find the mean and variance of the random variable $Z = 5X + 3$.

x	0	1	2	3
$P(X=x)$	0.4	0.3	0.2	0.1

$$E(X) = 0(0.4) + 1(0.3) + 2(0.2) + 3(0.1) = \underline{1}$$

$$E(X^2) = 0^2(0.4) + 1^2(0.3) + 2^2(0.2) + 3^2(0.1) = 2$$

$$\Rightarrow \sigma_X^2 = E(X^2) - E(X)^2 = 2 - 1^2 = \underline{1}$$

$$Z = 5X + 3$$

$$E(Z) = E(5X+3) = 5E(X) + 3 = 5(1) + 3 = \underline{8}$$

$$\sigma_Z^2 = \sigma_{5X+3}^2 = (5)^2\sigma_X^2 = 25(1) = \underline{25}$$

2. If a random variable X is defined such that $E[(X-1)^2] = 10$ and $E[(X-2)^2] = 6$, find μ and σ^2.

$$E[(X-1)^2] = E[X^2 - 2X + 1] = E(X^2) - 2E(X) + 1 = 10$$

$$E[(X-2)^2] = E[X^2 - 4X + 4] = E(X^2) - 4E(X) + 4 = 6$$

$$\Rightarrow E(X^2) = 2E(X) + 9$$

$$\Rightarrow 2E(X) + 9 - 4E(X) + 4 = 6$$

$$-2E(X) = -7 \Rightarrow \underline{E(X) = 7/2}$$

$$\Rightarrow E(X^2) = 2(7/2) + 9$$

$$= 16$$

$$\Rightarrow \sigma^2 = E(X^2) - E(X)^2 = 16 - (7/2)^2 = \underline{3.75}$$

3. If X and Y are independent random variables with variances $\sigma_X^2 = 5$ and $\sigma_Y^2 = 3$, find the variance of the random variable $Z = -2X + 4Y - 3$.

$$\sigma_Z^2 = \sigma^2_{-2X+4Y-3} = \sigma^2_{-2X+4Y} = (-2)^2\sigma_X^2 + (4)^2\sigma_Y^2$$

$$= 4(5) + 16(3) = \underline{68}$$

4. Let X represent the number that occurs when a red die is tossed and Y denote the number that occurs when a green die is tossed. Find

Both are fair.

(a) $E(X+Y)$,

(b) $E(X-Y)$,

(c) $E(XY)$.

(a) $E(X+Y) = E(X) + E(Y) = 2E(X)$

$E(X) = 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6)$

$= 3.5$

$\Rightarrow E(X+Y) = \underline{7}$

(b) $E(X-Y) = E(X) - E(Y) = \underline{0}$

(c) $E(XY) = E(X)E(Y)$ (independent events)

$= (3.5)^2 = \underline{12.25}$

5. A random variable X has a mean $\mu = 10$ and a variance $\sigma^2 = 4$. Using Chebyshev's theorem, find

(a) $P(|X - 10| \geq 3)$,

(b) $P(|X - 10| < 3)$,

(c) $P(5 < X < 15)$,

(a) $P(|X-10| \geq 3) = 1 - P(|X-10| < 3)$

$P(|X-10| < 3) = P(-3 < X - 10 < 3)$

$= P(10 - 3 < X < 10 + 3)$

$= P(10 - (\frac{3}{2})(2) < X < 10 + (3/2)(2))$

$\Rightarrow k = 3/2 \Rightarrow$

$P(|X-10| < 3) \geq 1 - \frac{1}{(3/2)^2} = 1 - 4/9 = 5/9$

$\Rightarrow P(|X-10| \geq 3) = 1 - P(|X-10| < 3) \leq 1 - 5/9 = \frac{4}{9}$

$\Rightarrow P(|X-10| \geq 3) \leq 4/9$

(b) $P(|X-10| < 3) \geq 5/9$

5. A random variable X has a mean $\mu = 10$ and a variance $\sigma^2 = 4$. Using Chebyshev's theorem, find

(a) $P(|X - 10| \geq 3)$,

(b) $P(|X - 10| < 3)$,

(c) $P(5 < X < 15)$,

(c) $P(5 < X < 15)$

$= P(10 - 5 < X < 10 + 5)$

$k = 5/2$

$= P(\mu - (5/2)2 < X < \mu + (5/2)2)$

$\geq 1 - \frac{1}{(5/2)^2} = 1 - \frac{4}{25} = \frac{21}{25}$

$\Rightarrow P(5 < X < 15) \geq \frac{21}{25}$